Automated Behavioural Analysis in Parent-Infant Interactions using Multimodal-Large-Language Models

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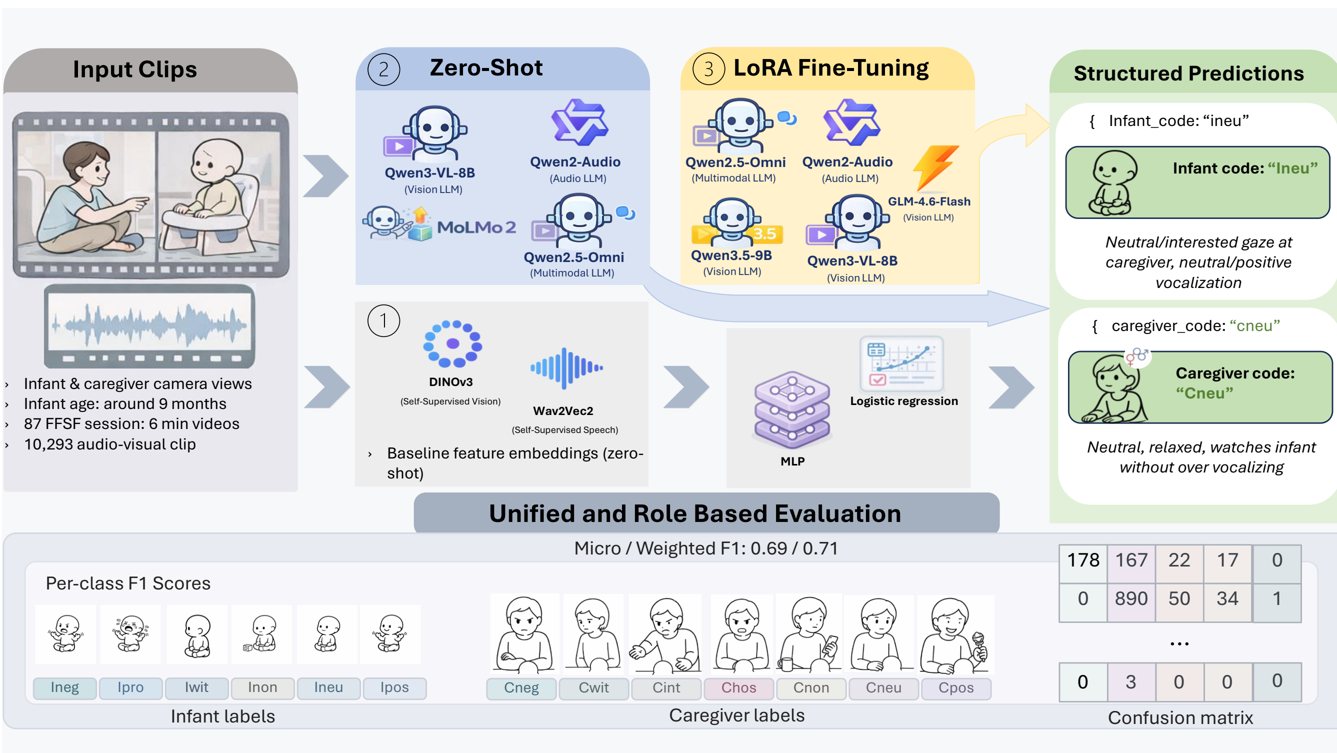

Figure 1: Overview of our process

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Abstract

Fine-grained behavioral coding of parent–infant interaction is central to developmental and clinical research, but manual annotation with schemes such as the Infant and Caregiver Engagement Phases–Revised (ICEP-R) is time-consuming, costly, and difficult to scale. This limitation is particularly important in longitudinal and clinically sensitive settings, where reliable analysis of early interaction is needed to study caregiver mental health, attachment, and infant socio-emotional development. In this work, we investigate whether recent multimodal foundation models and Multimodal Large Language Models (MLLMs) can support automated ICEP-R coding in the Face-to-Face Still-Face (FFSF) paradigm. We present a unified pipeline for video-only, audio-only, and audiovisual prediction, and compare three transfer routes: 1) frozen-feature downstream task classification baselines, 2) zero-shot prompting, and 3) parameter-efficient fine-tuning of multimodal models for infant and caregiver engagement prediction. We evaluate the approach on 87 FFSF sessions involving infants of approximately 9 months of age. Across settings, multimodal fusion consistently outperforms single-modality approaches, while infant engagement prediction remains more difficult than caregiver prediction due to subtler and noisier behavioral cues. Despite strong general-purpose multimodal capabilities, zero-shot MLLMs perform substantially worse than both classical baselines and fine-tuned models, indicating that domain adaptation is essential for this clinically specific task. The best performance is obtained with fine-tuned Qwen2.5-Omni-7B, which achieves weighted F1 scores of 0.657 for infant prediction and 0.692 for caregiver prediction, outperforming strong frozen-feature baselines based on DINOv3 and Wav2Vec 2.0. These findings suggest that parameter-efficient adaptation of multimodal large language models is a promising direction for scalable behavioral coding in developmental research, while highlighting persistent challenges in rare engagement states and subtle infant social signals.

CCS Concepts

• **Computing methodologies → Computer vision**; **Video analysis**; *Machine learning*; *Transfer learning*.

Keywords

Multimodal Learning; Parent-Infant Interaction; Behavioral Coding; ICEP-R; Multimodal Large Language Models; Vision-Language Models; Audio-Language Models; Automated Annotation

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1 Introduction

Early infant–caregiver interaction is a multidimensional social process central to developmental and clinical psychology research. Within the Face-to-Face Still-Face (FFSF) paradigm, the Infant-Caregiver Engagement Phases–Revised (ICEP-R) framework describes these dynamics at a fine temporal scale, capturing engagement, mismatch, repair, and co-regulation between interaction partners [25, 29]. Because variation in parent–infant interaction quality is linked to postpartum anxiety, antepartum emotional stress, and later psychobiological outcomes, detailed ICEP-R annotation carries clear clinical relevance [19, 26, 32].

At the same time, manual ICEP-R coding is costly and difficult to scale: specialized annotators are scarce, relevant signals are distributed across video and audio and span both interaction partners, and the available datasets are typically small and privacy-constrained. Reviews of parent–child interaction research highlight the continued reliance on expert observational coding and the limited standardization of computational support pipelines [2, 36], and recent survey work similarly calls for infant-specific perception tools, temporally grounded dyadic modeling, and shared evaluation settings [14].

Previous computational work shows that automated analysis is feasible, but much of it targets narrower subproblems or label-only prediction: mutual gaze and joint-attention detection [15], contact detection [8], head-movement analysis in still-face interactions [12], infant engagement prediction from facial images [10], and ICEP-R prediction from frozen audio–visual features with lightweight classifiers [31]. These studies recover useful behavioral structure but typically stop at label assignment, whereas expert coders interpret labels through modality-specific observations that are inspected, discussed, and sometimes revised. This motivates asking not only whether a model can predict the correct engagement code, but also whether it can provide a structured account of the cues underlying that prediction.

This question is timely given the rapid development of pretrained multimodal models. Self-supervised encoders such as DINOv2 [21] and DINOv3 [27] provide strong transferable visual representations, audio models such as w2v-BERT [7] provide comparably strong audio representations, and Multimodal Large Language Models (MLLMs) such as LLaVA [17], Qwen2-Audio [6], and Qwen2.5-Omni [33] now support instruction-based analysis of multimodal inputs. For behavioral coding, MLLMs are attractive because they can return both a predicted label and a short textual description of the cues treated as evidence, occupying a middle ground between pure label prediction and fully supervised explanation modeling; annotated datasets typically provide expert codes such as ICEP-R labels but not aligned expert-authored rationales explaining why a label was assigned [14].

In this study, we investigate automated ICEP-R micro-behavioral coding [25] as a multimodal structured prediction problem in a privacy-sensitive, resource-constrained setting. Under a shared task definition, clip inventory, split policy, prompt schema, and evaluation protocol, we compare three transfer routes: *frozen-feature transfer*, where the pretrained encoder stays fixed and only a shallow downstream classifier is trained; *zero-shot prompting*, where a multimodal instruction-following model is used without task-specific updates; and *parameter-efficient fine-tuning*, where a pretrained multimodal backbone is adapted with LoRA [13]. The primary evaluation target throughout is exact ICEP-R label prediction, enabling direct comparison with prior work. Additionally, model-generated rationales are analyzed as a preliminary qualitative indicator of explanation-bearing annotation support.

To our knowledge, this is the first study to apply MLLMs to fine-grained parent–infant behavioral coding; rather than a new architecture, it aims to give future work, including rare-class-aware training and temporally grounded modeling of dyadic co-regulation, a common empirical footing. In summary, this study has four main contributions. First, we provide a controlled comparison of zero-shot MLLMs, frozen-feature baselines, and fine-tuned MLLMs for ICEP-R coding in the FFSF paradigm. Second, we show that zero-shot models recover some behavioral structure but remain weak in exact label alignment, whereas fine-tuning yields clear improvements over strong frozen-feature baselines, particularly audiovisually. This limited zero-shot performance of frontier MLLMs zero-shot suggests general-purpose multimodal models are less readily usable off the shelf for clinical micro-coding than often assumed. Third, we analyze zero-shot MLLM rationales as a preliminary, non-quantitative diagnostic, showing these models often detect plausible micro-behaviors but misalign them with the ICEP-R ontology. Fourth, per-class analysis and confusion matrices show the main remaining bottleneck is the long-tail difficulty of rare behavioral states, not an absence of useful multimodal signals.

For reproducibility, we will make the feature extraction pipeline, prompt templates used for fine-tuning and evaluation, together with the corresponding codebase, publicly available in GitHub under the following link: anonymous link.

2 Background and Related Work

2.1 Methodological Background

Developmental and Clinical Foundations. The Face-to-Face Still-Face (FFSF) paradigm is a standard setting for studying early infant–caregiver interaction because it makes dyadic regulation, mismatch, and repair observable over time [18, 24, 29]. In a typical FFSF sequence, the caregiver transitions from normal interaction to a still-face episode and then to reunion, eliciting systematic changes in infant affect, attention, and regulatory behavior [18]. In this work, we study FFSF interactions and use the Infant-Caregiver Engagement Phases–Revised (ICEP-R) framework as the prediction target, since it captures infant and caregiver engagement at a temporally fine-grained and theoretically meaningful level [25].

From Manual Coding to Computational Support. Observational coding remains central in research on parent–infant and parent–child interaction, but in practice it involves more than assigning a class label. Expert annotators also rely on modality-specific cues to justify a label and to determine whether a prediction is reliable enough to be accepted or revised [9, 28]. This motivates computational support approaches that are compatible with selective human correction rather than full replacement of expert judgment. Annotation tools introduced an expert-in-the-loop workflow in which models propose labels for human revision [3], while Carcone et al. [5] demonstrated that expert-coded interaction data can also serve directly as supervision targets for automatic coding systems. Motivated by prior work, we adopt this perspective and treat ICEP-R prediction as structured label prediction within a workflow that remains compatible with expert review.

Transfer Routes and Multimodal Rationale Generation. In this work, we compare three transfer routes for behavioral coding: frozen-feature transfer, zero-shot prompting, and parameter-efficient fine-tuning with LoRA [13]. This distinction is particularly relevant for modern multimodal models, since MLLMs can produce not only predicted labels but also short structured descriptions of the cues underlying those predictions. In the present work, we treat such textual outputs as analyzable model hypotheses rather than as automatically faithful explanations [34].

2.2 Related Work on Interaction Analysis

Benchmarks for Adult Dyadic Interaction. Research on adult dyadic interaction has benefited from more standardized benchmark infrastructure than infant–caregiver interaction research. NoXi [4] and NoXi+J [11] provide multimodal datasets for engagement analysis in novice–expert interaction, while the MultiMediate challenges extend this toward multi-domain and cross-cultural generalization [20, 30]. Multimodal cues are clearly useful, but robust generalization remains difficult under variation in language, culture, task, and recording conditions.

Research on Infant–Caregiver Interaction. In the infant–caregiver setting, prior work is more fragmented and typically focuses on narrower subproblems: mutual gaze and joint attention [15], parent–infant contact [8], head-movement dynamics in still-face interactions [12], infant engagement prediction from facial images [10], and ICEP-R micro-behavior label prediction using PCA-reduced frozen audio–visual foundation-model embeddings with Bi-LSTM downstream classifiers [31]. Overall, this literature has mainly relied on tool-derived cues, task-specific supervised models, or frozen-feature transfer pipelines; instruction-based multimodal analysis remains largely underexplored in this domain.

Gap Addressed in this Work. Taken together, the existing literature provides valuable building blocks but offers limited controlled comparison across transfer routes for ICEP-R coding. We address this gap through a shared task definition, split policy, and evaluation protocol that allow direct comparison between frozen-feature transfer, zero-shot prompting, and parameter-efficient fine-tuning. In addition to exact label prediction, we also analyze model-generated rationales qualitatively, since no gold-standard expert rationale annotations are currently available.

3 Audio-Video Raw Dataset

We use an ICEP-R-coded corpus of parent–infant interaction collected in two developmental research labs in Germany. The full corpus contains 108 recorded sessions; after excluding sessions with missing or unusable audio/video data and technically compromised recordings, 87 sessions remain for analysis. The average age of the children in the recordings is approximately 9 months. All ICEP-R labels were assigned by a single trained expert annotator following the project annotation protocol.

The visual data consist of split-screen video recordings that combine two synchronized camera views into a single frame, with a resolution of 1920×1080 pixels and a frame rate of 25 FPS. The two views typically show the upper body and face of each interaction

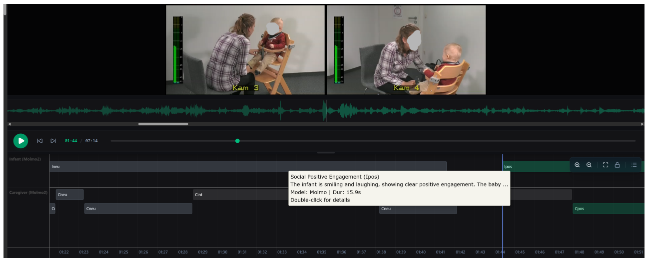

Figure 2: Example FFSF interaction segment with split-screen camera views (top) and model-predicted infant and caregiver annotations over time (bottom).

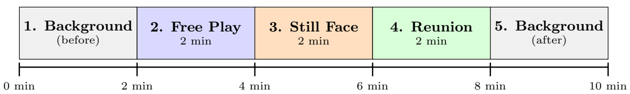

Figure 3: FFSF Timeline

partner from complementary angles, including side and over-the-shoulder perspectives as shown in Figure 2. Audio was recorded at 48 kHz using a centrally placed microphone between the interaction partners and resampled to 16 kHz for the experiments.

Session-Level Splitting. Based on the expert annotations for the infant and caregiver within each FFSF phase, as illustrated in Figure 3, we segment the sessions into 10,293 clips. Because segmentation follows behavioral state boundaries, clip duration is variable and ranges from 0.5 s to more than 60 s. Unless stated otherwise, experiments are conducted in a no-background setting that excludes clips labeled as background (bg), including footage recorded before and after the 6-minute FFSF interaction. Under this setting, the audiovisual dataset contains 9,267 active clips, of which 7,342 belong to the train-development partition and 1,925 belong to the held-out test partition.

To prevent data leakage across partitions, all splits are performed at the *session level* rather than the clip level, so that clips from a given session remain in the same fold or test partition. We use a separate held-out test partition and perform five-fold cross-validation on the remaining train-development sessions, stratified by recording site and coarse duration statistics to preserve comparable data composition across partitions.

4 Methodology

We formulate automated ICEP-R coding as a multimodal structured prediction problem over short parent–infant interaction clips, comparing three transfer routes under a shared experimental setup: zero-shot MLLMs, frozen-feature baselines built on pretrained visual and audio encoders, and parameter-efficiently fine-tuned MLLMs. Across all experiments, we keep the clip inventory, split policy, target semantics, and evaluation protocol fixed to enable controlled comparison.

Task Formulation. Each sample is a short interaction clip $x = (x_v, x_a)$, where video x_v, audio x_a, or both may be available. The target is the infant engagement code y_i, the caregiver code y_c, or both jointly, yielding infant-only, caregiver-only, and joint settings; we vary the input modality (video-only, audio-only, audiovisual) and, in some variants, whether background (bg) is included in the active label space.

Table 1: Compact summary of the ICEP-R label space used in this work. Full label definitions are provided in the supplementary material (Sec. G).

Role	Label groups in experiments
Infant	**`ineg`**/**`ipro`**: negative or protesting; **`iwit`**: withdrawn; **`inon`**: object/environment engagement; **`ineu`**: neutral social monitoring; **`ipos`**: positive social engagement; **`islp`**: sleep; **`iusc`**: unscorable; **`bg`**: background (variant-dependent).
Caregiver	**`cneg`**/**`cwit`**/**`chos`**: negative, withdrawn, or hostile; **`cint`**: intrusive; **`cnon`**: non-infant-focused; **`cneu`**: neutral monitoring; **`cpvc`**: monitoring with positive vocalization; **`cpos`**: positive social engagement; **`ctch`**: rough touch; **`cusc`**: unscorable; **`bg`**: background (variant-dependent).

The ICEP-R labels capture micro-temporal engagement states for both members of the dyad. For infants, the label space distinguishes negative or protesting states, withdrawal, object/environment engagement, neutral social monitoring, positive social engagement, and special states such as sleep, unscorable segments, and optional background. For caregivers, the labels distinguish negative, withdrawn, intrusive, hostile, or non-infant-focused behavior, neutral and positively vocalized monitoring, positive social engagement, rough touch, unscorable segments, and optional background. A compact summary is given in Table 1; full definitions are provided in the supplementary material (Sec. G).

All model families are evaluated against the same categorical targets. Generative systems produce a constrained structured output in JSON format, i.e., a machine-readable key–value representation, containing fields such as `infant_code`, `caregiver_code`, `infant_description`, and `caregiver_description`. These outputs are then parsed, mapped back to the active ICEP-R label space, and evaluated against the ground-truth labels, as illustrated in Figure 2, whereas frozen-feature baselines predict the corresponding labels directly.

Role. We consider three role settings: infant-only prediction, caregiver-only prediction, and joint prediction of both roles. The role-specific settings are motivated by the annotation procedure itself, since infant and caregiver states are coded separately rather than in a single joint annotation step; we test whether this separation also benefits machine prediction. The joint setting, in turn, tests whether shared multimodal context across the dyad improves prediction.

Shared Data Interface. All experiments are derived from a common multimodal clip inventory containing aligned media references, prompts, and gold annotations; experimental variants select the active modality, target role, and label inventory while preserving clip boundaries and gold labels. Generative-model prompts cast the system as an expert ICEP-R coder, specify the valid label set for the active setting, and require *JSON only* output, with the joint setting returning infant and caregiver codes plus short textual descriptions and single-role settings returning only the active role's fields. Full record structure and prompt templates are provided in the supplementary material (Secs. D, D.4, and C.1).

Model Families. Figure 1 illustrates the three transfer routes and corresponding models. In the frozen-feature route, pretrained visual and audio encoders stay fixed and produce clip-level representations classified by shallow downstream models. In the zero-shot route, multimodal instruction-following models are prompted directly with the shared ICEP-R task schema without task-specific updates. In the fine-tuning route, open-weight multimodal backbones are adapted with LoRA [13] to emit the target structured JSON output directly. The frozen-feature baselines use DINOv3 [27] visual and wav2vec 2.0 Large [1] audio features with logistic regression or a lightweight MLP; zero-shot and fine-tuned experiments were performed using LLaMA-Factory [35] and they cover video-only, audio-only, and audiovisual models from the Molmo, Qwen, and GLM families. Full model and hyperparameter details can be found in the supplementary material (Sec. A).

Frozen-Feature Baseline. To maintain comparability with prior ICEP-R work, we follow the general transfer setting of [31], who use frozen audio–visual foundation-model features with supervised downstream prediction, but differ in the downstream learner. In this article, we use shallow classifiers on pooled clip-level representations rather than a Bi-LSTM-style temporal model, closer to standard frozen-feature evaluation practice [22, 27]. Additional implementation details are in the supplementary material (Sec. B).

Zero-Shot and Fine-Tuned MLLMs. In the zero-shot setting, models receive all task information through in-context instructions, including the active ICEP-R label definitions, output constraints, and validation rules, and are required to return predictions in the same constrained JSON format used across generative experiments. This allows us to assess how much ICEP-R-relevant behavioral structure can be recovered from pretrained multimodal instruction-following models without task-specific adaptation.

In the fine-tuned setting, multimodal backbones are adapted with LoRA so that they learn to emit the target ICEP-R schema directly from supervised examples. At inference time, decoding is deterministic. Generated outputs are post-processed by extracting the first valid JSON object and mapping the predicted fields back to the active ICEP-R label space, yielding $\hat{y}_i$ and/or $\hat{y}_c$. Parse errors and invalid outputs are tracked separately from ordinary classification errors.

5 Results

We report results for three transfer routes as illustrated in Figure 1. The quantitative comparison remains centered on expert-provided ICEP-R labels, since the current corpus contains categorical annotations but not expert-authored explanations aligned with individual label decisions. F1 score is therefore the primary metric, in line with prior work [31]. We further complement aggregate scores with per-class analysis, confusion matrices, and qualitative inspection of the descriptive outputs returned by the zero-shot MLLMs. These descriptions are analyzed only qualitatively, since the absence of expert-authored references precludes evaluation with ROUGE [16], BLEU [23], or related semantic-overlap metrics.

Zero-Shot MLLMs. The zero-shot MLLMs in Figure 4 provide a useful lower bound on how much ICEP-R structure is accessible from generic multimodal instruction-following pretraining alone. All zero-shot systems remain far below the frozen-feature baselines and fine-tuned models in weighted F1. However, modality differences are noticeable: on the one hand, the strongest infant score is achieved by the audiovisual **Qwen2.5-Omni-7B** model (0.272), followed by vision-only **Molmo2** (0.206); on the other hand, the strongest caregiver score is obtained with the visual **Molmo2** model (0.195), followed by **Qwen3-VL-8B** (0.153) and audiovisual **Qwen2.5-Omni-7B** (0.126). The audio-only zero-shot model is effectively non-functional here.

The confusion matrices in the supplementary material (Figure S1) show that this miss-classification pattern is not merely low confidence: zero-shot models frequently fail to form usable class boundaries and collapse onto one majority class or a small label subset. Therefore, generic multimodal instruction models recover coarse interactional structure, but not the fine-grained clinically meaningful decision surfaces that ICEP-R coding requires, motivating the fine-tuning experiments.

Qualitative diagnosis of zero-shot low performances. Table 2 complements the zero-shot results with a small set of hand-picked examples illustrating why zero-shot MLLMs perform poorly, read together with the confusion matrices in the supplementary material (Figure S1). This analysis is diagnostic rather than evaluative: with no gold-standard rationale annotations, we treat the examples as a preliminary observation about failure modes, not a quantitatively validated finding.

Two patterns stand out. First, incorrect predictions often contain semantically relevant evidence: the Molmo2 example identifies sustained physical contact and repositioning but maps these cues to `cint` rather than the target `cpvc`—not an arbitrary error, but the wrong decision boundary applied to plausible cues. Second, the qualitative failures mirror the quantitative transfer limitations: the audio-only Qwen2-Audio example shows a domain-mismatch failure, where high-pitched infant vocalization is read as distress rather than positive or neutral pre-verbal vocalization, consistent with the weak zero-shot audio results and the model's negative-prediction bias, while the successful audiovisual Qwen2.5-Omni example shows the intended use case: the model predicts the correct caregiver label and aligns it with convergent visual and prosodic evidence.

Comparison across Frozen-Feature Baselines and Fine-Tuned MLLMs. As it can be observed in Figure 4 , the Frozen-Feature baselines establish a strong reference point: the best unimodal audio model reaches a weighted F1 of 0.462 (infant) / 0.555 (caregiver), the best unimodal video model reaches 0.540 / 0.522, and the fused audiovisual baseline improves further up to 0.569 / 0.631, confirming that multimodal fusion is already beneficial before model adaptation, surpassing the Zero-Shot multimodal model (Qwen2.5-Omni) in +29.70 and +50.50 percentage points, respectively.

Moving to the analysis of the Fine-Tuned MLLMs, it is noticeable that the W-F1 score increases substantially over this Frozen-Feature baseline. The strongest overall joint model is **Qwen2.5-Omni-7B**, which reaches 0.657 weighted F1 on infant labels and 0.692 on caregiver labels. Relative to the best frozen-feature fused baseline,

Table 2: Qualitative examples of zero-shot generated rationales. [AUD] and [VIS] denote audio and visual evidence referenced by the model. Examples are hand-selected from different sessions to illustrate a correct prediction and two representative error modes. This analysis is diagnostic rather than evaluative (Section 6). The complete task-specific prompts are provided in the supplementary material (Sec. C).

Model & Modality	Metadata (Length)	Target vs. Predicted Code	Outcome Category	Generated Rationale	Our Interpretation
Molmo2-8B *(Vision-Only)*	Session_003 *Clip dur: 14.5s*	**Target:** `cpvc` **Pred:** `cint`	**Semantically Plausible Error** (Semantic Drift)	*'The caregiver uses frequent gestures and maintains close physical contact with the baby, adjusting her position to stay near.'*	The model captures the relevant physical contact and repositioning cues, but maps them to `cint` rather than the milder threshold associated with `cpvc`.
Qwen2-Audio *(Audio-Only)*	Session_079 *Clip dur: 9.8s*	**Target:** `ipos` **Pred:** `ineg`	**Distribution-Mismatch Error**	*'The infant emits a series of high-pitched, unintelligible vocalizations, indicating potential fussiness or mild distress.'*	The model isolates high pitch correctly, but interprets pre-verbal infant vocalization through an adult-centered distress prior and collapses the case into `ineg`.
Qwen2.5-Omni *(Audio-Visual)*	Session_001 *Clip dur: 5.6s*	**Target:** `cpos` **Pred:** `cpos`	**Correct Prediction** (Multimodal Agreement)	*'[VIS]: The mother is smiling widely... [AUD]: The mother uses a highly exaggerated, sing-song voice characteristic of motherese...'*	Visual and prosodic evidence align cleanly with `cpos`, illustrating the intended multimodal behavior of the model.

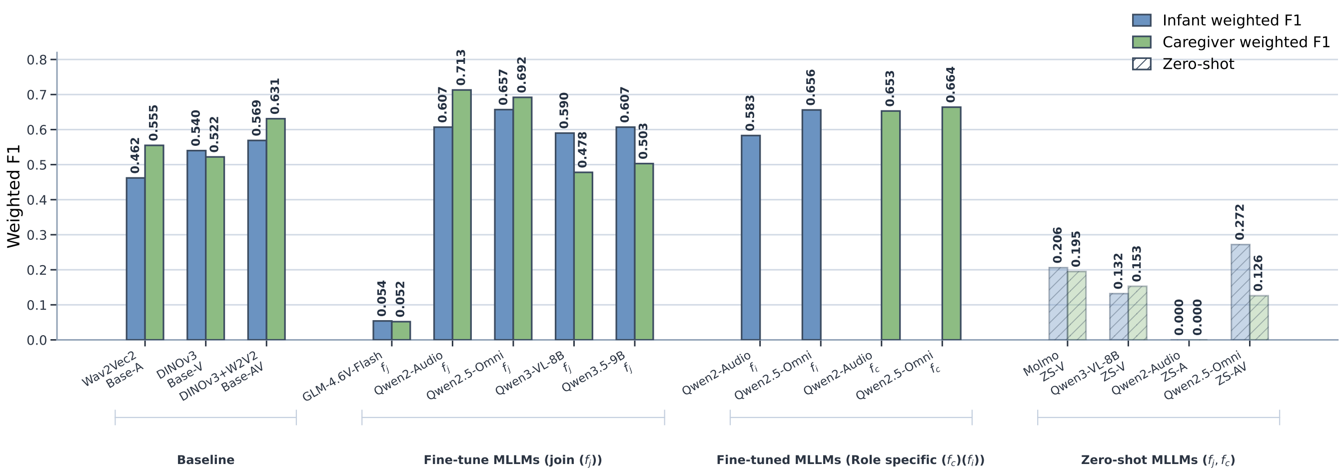

Figure 4: Weighted F1 across frozen-feature baselines, fine-tuned MLLMs, role-specific (fine-tuned) MLLMs, and zero-shot MLLMs for ICEP-R coding. Blue: infant weighted F1; green: caregiver weighted F1. Joint prediction uses $f_j : x \mapsto (y_i, y_c)$; role-specific prediction uses $f_i : x \mapsto y_i$ and $f_c : x \mapsto y_c$.

this corresponds to gains of roughly 9 percentage points on infant prediction and 6 percentage points on caregiver prediction. The strongest caregiver score in the overview is achieved by the joint **Qwen2-Audio-7B** model, which reaches 0.713, suggesting that acoustic information is especially informative for caregiver-side coding, while its infant performance (0.607) remains below the joint Omni model, indicating that the audiovisual model is more balanced across both roles.

The comparison among Fine-Tuned MLLMs also reveals a clear modality effect: video-only models improve over the Frozen-Feature baselines but underperform compared to the joint audiovisual model, especially on caregiver prediction (**Qwen3-VL-8B**: 0.590/0.478; **Qwen3.5-9B**: 0.607/0.503), suggesting visual cues alone cannot match the stronger multimodal and audio-driven systems here.

The role-specific models, motivated by the fact that infant and caregiver annotations were produced independently, do not consistently outperform the corresponding joint models: **Qwen2.5-Omni-7B** reaches 0.656 (infant-only) and 0.664 (caregiver-only), both below or tied with the joint results of 0.657 and 0.692, and **Qwen2-Audio-7B** shows a similar pattern. These differences are small (1–3 points) relative to the gaps separating zero-shot, frozen-feature, and fine-tuned models, and untested for statistical significance; we therefore do not find evidence that role-specialized adaptation outperforms joint supervision in this dataset.

Finally, **GLM-4.6V-Flash** remains near chance level (0.054 infant, 0.052 caregiver), but this does not necessarily indicate an absence of multimodal reasoning ability: raw outputs show the model often defaults to an open-ended deliberative style (e.g., "let's tackle this step by step ... is the caregiver using physical control? or `cpos`?") and fails to terminate in a valid task-constrained JSON

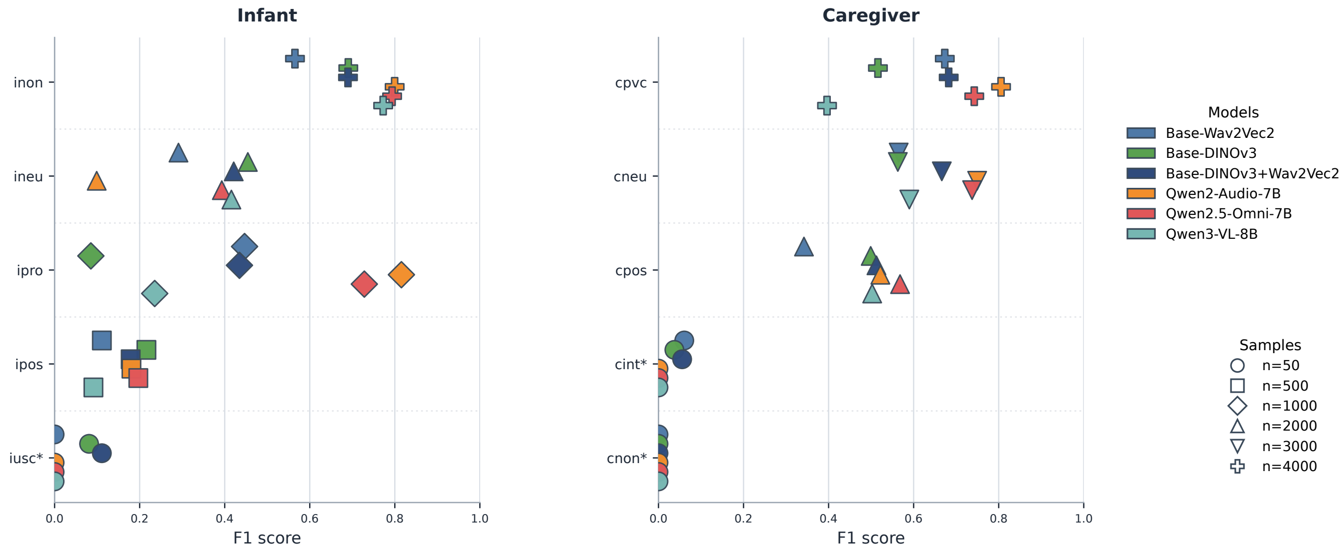

Figure 5: Per-class F1 for infant and caregiver coding. Colors indicate model identity; marker shape encodes approximate class sample count. Improvements concentrate in common classes (`inon`, `ipro`, `cneu`, `cpvc`); rare classes (asterisks: fewer than 50 examples) remain difficult across all methods.

prediction. Its weakness is therefore an inability to output parsable JSON reliably; representative raw outputs are in the supplementary material (Sec. E).

Per-class comparison. Aggregate weighted F1 does not show where the gains arise. Figure 5 therefore reports per-class F1 for the strongest frozen-feature baselines and fine-tuned models, while also indicating approximate class support. This per-class view, together with the confusion matrices in the supplementary material (Figure S1), characterizes which behavioral states are learned reliably; it complements rather than replaces a full deployment evaluation, which would additionally require joint infant–caregiver correctness and uncertainty estimation for human-in-the-loop use (Section 6).

Three patterns are most relevant. First, class support is strongly associated with learnability: F1 is highest for well-represented classes such as infant `inon` and caregiver `cneu`/`cpvc`. Second, fine-tuned MLLMs close the gap on moderately supported classes more effectively than frozen-feature baselines; for infant `ipro`, frozen features remain substantially weaker, whereas audio and audiovisual MLLMs learn this class much more effectively, suggesting that task-specific adaptation improves representation learning beyond fixed pooled features. Third, the lowest-support classes remain unresolved across all methods: infant `iusc` and caregiver `cint`/`cnon` stay near zero for most models, indicating a data-representation limitation rather than a model-capacity one.

The figure also highlights a modality-specific tradeoff. **Qwen2-Audio-7B** is particularly strong on several caregiver classes, consistent with the caregiver-side gains in the aggregate results, whereas **Qwen2.5-Omni-7B** provides the strongest overall balance by improving both infant and caregiver classes simultaneously.

Summary of findings. Taken together, four conclusions emerge. First, zero-shot prompting is a useful reference point but not sufficient for reliable ICEP-R coding. Second, multimodality matters: DINOv3+wav2vec 2.0 yields the strongest frozen-feature model, and the joint audiovisual Omni model gives the strongest balance among fine-tuned models. Third, fine-tuning clearly advantages frozen-feature classification, with gains concentrated in frequent, behaviorally central classes; Figure S1 in the supplementary material shows these correspond to sharper class separation, not just higher aggregate scores, while residual errors stay structured (e.g., `ineu`, `ipos`, `iusc`, `iwit` absorbed into `inon`; `cint`/`cnon` mapped to `cneu`/`cpvc`). Fourth, the main unresolved challenge is class imbalance rather than an absence of recoverable multimodal signal. Beyond label prediction, the qualitative outputs suggest MLLMs can surface candidate evidence for their decisions, motivating future work on expert-grounded explanation supervision.

6 Discussion

Fine-tuning aligns representations, not just perception. The comparison across zero-shot, frozen-feature, and fine-tuned models suggests that the central challenge is not only perceptual access to the signal, but alignment to the ICEP-R coding ontology. Zero-shot models recover limited behavioral structure and occasionally plausible cue-level descriptions, but do not map multimodal evidence reliably onto the target label space; frozen-feature baselines improve substantially, showing that generic pretrained representations already encode useful information, and fine-tuning improves further by aligning those representations with a fine-grained clinical coding scheme rather than by improving basic perception alone. The modality pattern reinforces this: the audiovisual Omni model gives the strongest overall balance, while the strongest caregiver score comes from the audio-only Qwen2-Audio model, suggesting caregiver states are often carried by prosody and vocal style whereas infant coding depends more on visual cues such as gaze, posture, and social orientation—multimodality matters, but not symmetrically across roles.

The main unresolved challenge is the long tail. The per-class and confusion analyses point to the same conclusion. Gains concentrate in frequent, behaviorally central classes, whereas rare states—infant `iusc`, caregiver `cint`/`cnon`—remain difficult across all transfer routes and are often absorbed into nearby majority categories. The bottleneck is therefore not whether the models detect interactional signal at all, but whether they can separate weakly represented, subtle states from neighboring high-frequency classes; future gains are more likely to come from better coverage of underrepresented classes, targeted rebalancing, or rare-class-aware objectives than from scaling the backbone further. Role-specific fine-tuning does not add a consistent advantage over joint training either: the role-specific models remain competitive but do not clearly outperform the joint models, and given that the observed differences are small (1–3 points) and untested for significance, we treat this as a non-inferiority result rather than evidence that joint supervision is strictly better. Since infant and caregiver behavior are coupled components of the same dyadic process, joint supervision remains a reasonable default for future work.

Beyond label prediction: toward expert-in-the-loop annotation support. A central motivation of this paper is to position multimodal generative models slightly beyond label prediction alone. Prior ICEP-R automation work focused on recovering the correct categorical label from frozen pretrained features with downstream classifiers; the MLLMs studied here retain label prediction as the primary benchmark but also expose a second capability, short textual descriptions of the cues associated with a decision, used here diagnostically rather than as a scored target since no expert-authored explanations are available. Even so, they suggest weak zero-shot performance often reflects ontology misalignment rather than complete perceptual failure, and point toward annotation-support tools rather than full replacements for expert coding: the strongest models seem reliable enough for common interaction states, while rare or clinically subtle cases should stay with expert review. A concrete next step is collecting expert-grounded explanation traces, by asking annotators to justify decisions directly or revise model-generated descriptions.

Limitations. All ICEP-R labels were produced by a single trained annotator; we did not collect a second independent coding pass, so no inter-rater reliability statistic (e.g., Cohen's or weighted κ) is available, and results should be read as agreement with one expert's coding rather than a consensus ground truth. The dataset is relatively small and imbalanced, the prediction task is clip-level rather than fully temporal even though ICEP-R is defined by temporal dynamics such as mismatch, repair, and co-regulation, and the descriptive MLLM outputs cannot yet be evaluated quantitatively for the same reason. We also do not systematically analyze confidence calibration or robustness to distribution shift, consistent with a controlled transfer-route comparison rather than a deployment-readiness study, so the present results should not be read as evidence of calibrated or robust clinical behavior; the zero-shot comparison likewise serves as a transfer baseline rather than a perfectly matched benchmark. These limitations define the scope of the current contribution rather than changing its overall picture. Representative training runs required single-digit to low double-digit GPU-hours (supplementary material, Sec. H), reported as a practical reproducibility summary rather than a full environmental accounting.

7 Conclusion

This paper studied automated ICEP-R coding as a controlled comparison of three transfer routes: zero-shot prompting, frozen-feature transfer, and parameter-efficient fine-tuning. Under a shared task definition, split policy, and evaluation protocol, parameter-efficient multimodal fine-tuning clearly outperforms both zero-shot prompting and frozen-feature classification, while frozen audiovisual features already provide a strong baseline; the strongest models are audio or audiovisual, with caregiver coding benefiting particularly from acoustic information. More broadly, multimodal behavioral coding should not be framed only as label recovery: the generative models studied here also return cue-level audiovisual descriptions of their decisions, even though these cannot yet be evaluated against expert-authored explanations. The contribution is therefore twofold, a controlled benchmark of transfer routes for privacy-sensitive ICEP-R prediction and a preliminary step toward more transparent, expert-interactive annotation support, with next steps including rare-class-aware and temporally grounded modeling of dyadic co-regulation, and collecting second-annotator and expert-grounded explanation data to evaluate both inter-rater reliability and explanation quality.

Safe and Responsible Innovation Statement

This work operates on privacy-sensitive parent–infant video and audio recorded under informed consent for research use; raw recordings are not redistributed. Fine-tuned LoRA adapters can memorise training examples and are susceptible to membership-inference and model-inversion attacks, so any release of trained weights would require institutional data-protection review rather than being assumed safe. We describe the training and evaluation pipeline in enough detail for other developmental-research groups to reproduce the methodology on their own protected datasets. The systems described are research aids for expert-in-the-loop annotation, not diagnostic tools: they should not replace clinical judgement, and deployment outside the training distribution (different ages, sites, languages, or paradigms) needs additional validation before clinical or policy use. The authors used generative AI tools (ChatGPT, Claude) to assist with software development and text editing; all scientific content, analyses, and conclusions remain the authors' sole responsibility.

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